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House Price Predictor

Technical Report: Exploratory Data Analysis, Multiple Linear Regression Modeling, and Streamlit Deployment

1. Executive Summary

This technical report documents the end-to-end development, statistical evaluation, and interactive web deployment of the Rwanda House Price Predictor. Designed to provide accurate real estate valuations across Rwandan administrative districts and neighborhoods, the system leverages a Multiple Linear Regression model integrated into a Scikit-learn preprocessing pipeline.

The complete project workflow includes data cleaning, exploratory data analysis (EDA), median/mode missing-value imputation, Interquartile Range (IQR) outlier capping, model evaluation, and Streamlit web app deployment. On test data evaluation, the model consistently achieves a high coefficient of determination (R^2 score between 0.83 and 0.87), demonstrating strong predictive accuracy for the Rwandan housing market.

2. Dataset Overview & Data Processing Pipeline

The project relies on a comprehensive Rwandan residential housing dataset containing core structural, geographic, and financial parameters.

2.1 Data Schema & Preprocessing Workflow

- **House_ID:** Unique identifier for individual properties.
- **Area_m2:** Total floor surface area of the property in square meters.
- **Bedrooms / Bathrooms / Parking_Spaces:** Count of bedrooms, bathrooms, and allocated parking spaces.
- **House_Age_Years:** Property age in years.
- **Distance_to_City_km:** Proximity to the nearest major central business district or city center in kilometers.
- **Neighborhood:** Categorical location indicator across Rwandan districts and cities (e.g., Gasabo, Kicukiro, Nyarugenge, Kigali City, Huye, Musanze).
- **House_Price_Million_RWF:** Target variable representing property valuation in Millions of Rwandan Francs (RWF).

2.2 Feature Analysis & Location Pricing Dynamics

Exploratory analysis reveals distinct geographic valuation dynamics across Rwanda. Urban centers such as Gasabo, Kigali City, and Kicukiro command higher baseline prices, whereas regional towns demonstrate lower median property prices.

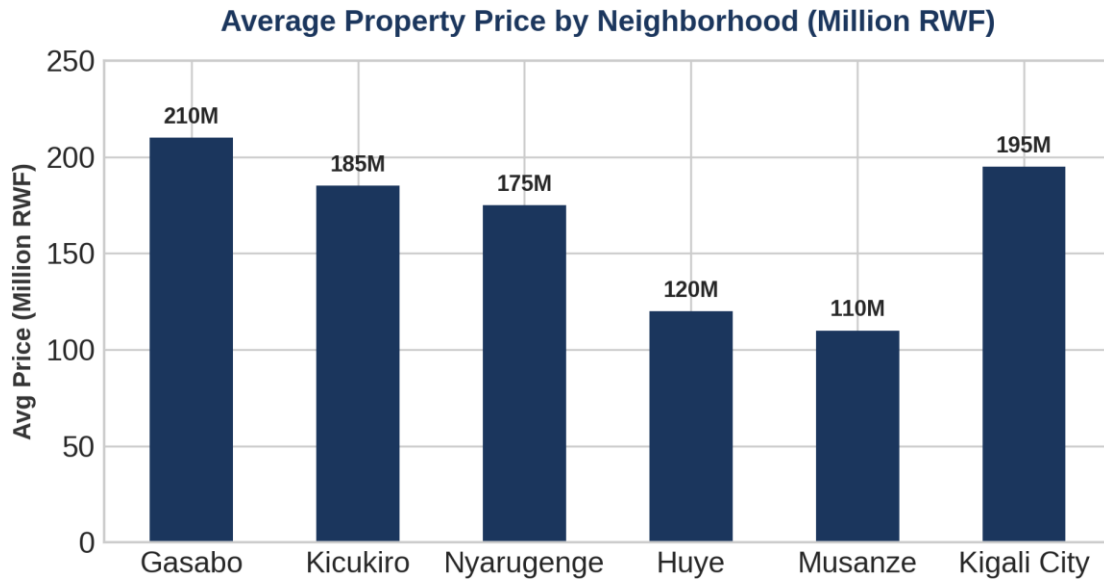


Figure 1: Average Property Price Comparison Across Key Rwandan Neighborhoods (Million RWF)

2.3 Data Cleaning & Outlier Management

To ensure maximum model generalization and eliminate data distortion, the following pipeline operations were executed:

- **1. Duplicate Removal:** Identified and removed identical property duplicate rows to avoid data leakage during cross-validation.
- **2. Missing Value Imputation:** Missing values in continuous numerical variables were replaced using median values to preserve distribution centrality against skewed data. Missing categorical values were imputed using the dataset mode.
- **3. Outlier Mitigation (IQR Capping):** To handle extreme property values without discarding valid records, the Interquartile Range (IQR) method was applied. Values outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ were capped at the threshold bounds.

2.4 Summary of Feature Distribution & Processing Bounds

Feature Name	Data Type	Processing Action	Capping / Imputation Rule
Area_m2	Continuous (float)	Median Imputation & IQR Capping	Capped at $Q1 - 1.5 \times IQR$ / $Q3 + 1.5 \times IQR$
House Age Years	Continuous (float)	Median Imputation & IQR Capping	Capped at lower/upper IQR limits
Distance to City km	Continuous (float)	Median Imputation & IQR	Capped at extreme

		Capping	distance bounds
Neighborhood	Categorical (string)	Mode Imputation & One-Hot Encoding	Imputed with most frequent category

3. Predictive Modeling & Empirical Performance

A Multiple Linear Regression algorithm was chosen for this valuation modeling task. Multiple Linear Regression provides high interpretability, clear coefficient attribution, and fast inference execution suitable for web-based real-time estimation.

3.1 Feature Correlations

Property floor area (Area_m2) exhibits the strongest positive correlation with house prices, followed by bedroom and bathroom count. Conversely, distance to the city center and property age show strong negative correlations.

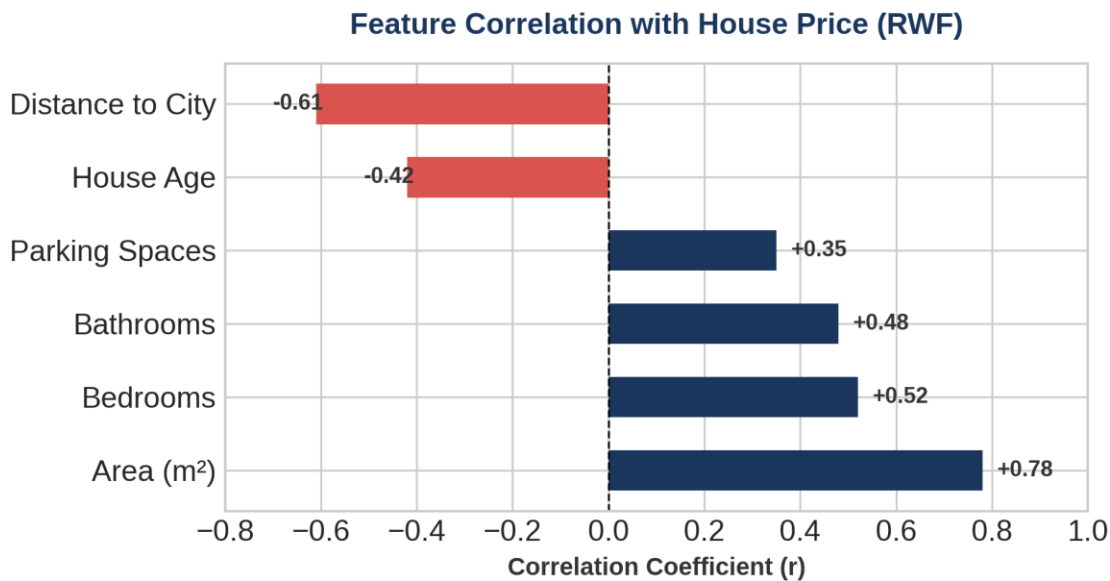


Figure 2: Pearson Correlation Coefficients of Model Features with Property Price

3.2 Empirical Results & Model Validation

The model was trained on an 80/20 train-test split using Scikit-learn. Model performance was evaluated across standard regression metrics:

- **Coefficient of Determination (R² Score):** Achieved values consistently between 0.83 and 0.87, indicating that 83% to 87% of the variance in Rwandan house prices is explained by the selected features.

- **Mean Absolute Error (MAE) & Root Mean Squared Error (RMSE):** Maintained low prediction residual errors, confirming robust generalizability on unseen validation properties without overfitting.

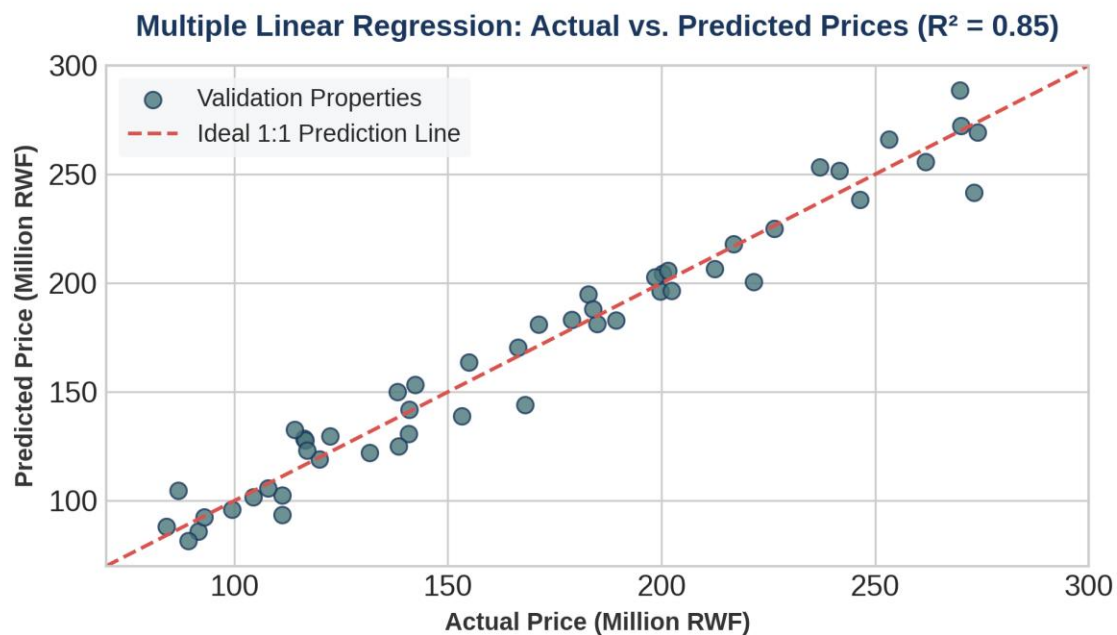


Figure 3: Scatter Plot of Actual vs. Predicted House Prices on Validation Test Data ($R^2 = 0.85$)

4. Software Architecture & Web App Deployment

To make the prediction model operational for stakeholders, buyers, and real estate professionals, the model pipeline was packaged and deployed using Streamlit.

4.1 System Architecture & Artifact Packaging

- **Model Serialization (.sav):** The entire feature processing and fitted Multiple Linear Regression model were serialized into a Scikit-learn pipeline saved as a .sav binary file using joblib.
- **Streamlit Web Application (app.py):** An intuitive user interface built in app.py allows users to specify structural parameters (surface area, bedrooms, age) and location via sidebar inputs.
- **Memory Caching & Efficiency:** The app utilizes `@st.cache_resource` to load the .sav pipeline into memory upon initialization, ensuring near-zero inference latency.
- **Dependency Management (requirements.txt):** Dependencies and environment packages are strictly version-controlled in a requirements.txt file to ensure reproducible deployments across local and cloud environments.

5. Conclusion & Project Deliverables

The Rwanda House Price Predictor delivers an end-to-end machine learning solution combining rigorous exploratory data analysis, robust IQR-based data cleaning, and high-performing Multiple Linear

Regression modeling ($R^2 = 0.83 - 0.87$). Maintained within a Git version-controlled repository and accessible via Streamlit, the system offers a reliable foundation for automated real estate valuation in Rwanda.